Ayush Kumar Singh

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Hostel-13, BIT Mesra, Ranchi (India) - 835215

Education

- **Birla Institute of Technology, Mesra** *Sept 2023 – Sept 2028*
  Integrated M.Sc. in Physics (Minor in Mathematics & Computing) Ranchi, India
  - CGPA: 8.72 / 10.0 (Branch Topper)
  - 4th Semester SGPA: 9.44

  Relevant Coursework: Classical Mechanics, Quantum Mechanics, Mathematical Physics, Computational Physics, Statistical Physics, Thermal Physics, Waves and Optics, General Relativity.

Publications & Conference Presentations

- ***Journal of Astrophysics and Astronomy* (Springer)** *3rd author reviewing draft*
  Manuscript in preparation
  - **Title:** UVIT Study of the Magellanic Clouds (U-SMAC) V. UV Stellar Populations in the Western Outskirts of the Small Magellanic Cloud
  - **Authors:** Ayush Singh, Sipra Hota, Prasanta Kumar Nayak, and Annapurni Subramaniam
- **A Decade of AstroSat Observations: Science Outcomes and Future Prospects** *January 2026*
  Host Organisation: U R Rao Satellite Centre (URSC), Bengaluru, India []
  - Presented a **poster** titled *"Exploring the Western Outskirts of the Small Magellanic Cloud: Tracing Foreground and SMC Stars with UVIT"*, co-authored with Sipra Hota, Prasanta Kumar Nayak, and Annapurni Subramaniam.

Research Experience

- **Indian Institute of Technology Hyderabad, India** *May 2026 – July 2026*
  Title: *Dark Matter and Beyond standard model Physics Research* **On-site**
  Advisor: Dr. Divya Sachdeva
  - Will work on dark matter research in the domain of particle physics under faculty guidance.
  - Will study fundamental concepts of particle physics, including Feynman diagrams and related theoretical frameworks.
  - Will explore introductory high-energy physics concepts relevant to dark matter phenomenology and interactions.
- **Indian Institute of Astrophysics, India** *Ongoing from May 2025*
  Title: *Disentangling Stellar Populations in the Western Halo of the SMC Using UVIT and Gaia DR3* **Hybrid**
  Advisors: Sipra Hota, Dr. Prasanta Kumar Nayak, and Prof. Annapurni Subramaniam
  - Conducted literature review and processed UVIT and Gaia DR3 data, including photometric calibration and generation of clean catalogs.
  - Applied advanced analysis techniques such as cross-matching, vector point diagrams, GMM, and KDE to study stellar kinematics and separate Milky Way foreground (including 47 Tuc) from SMC members.
  - Constructed UV/optical CMDs and color–color diagrams, performed isochrone fitting, and identified distinct stellar populations. *Manuscript in preparation*
- **Birla Institute of Technology, Mesra, India** *1 year: May 2024 – April 2025*
  Title: *Performance Analysis and Optimization of a Consumption-Scale Biogas Plant* **On-site**
  Advisor: Dr. Rajeev Kumar
  - Led the setup, data collection, and maintenance of a newly built biogas plant.
  - Analyzed system data to optimize efficiency, waste ratios, and mechanical performance.
  - Assisted Bachelor, Master, and Ph.D. students in data analysis and thesis preparation.
  - Developed Project that won an INR 10,000 award at the BIT NISHAN Business Competition.

Projects

- **Birla Institute of Technology, Mesra, India [🌐]** *From April 2026*
  Title: *Introduction to Bayesian Methods and Monte Carlo Techniques* **On-site**
  Advisor: Dr. Prakash Chandra
  ◦ Studying fundamentals of Bayesian methods and implementing basic probabilistic models for parameter estimation.
  ◦ Exploring Monte Carlo methods, focusing on sampling techniques as part of ongoing research training.

- **Birla Institute of Technology, Mesra, India** *3 months: April 2025 – June 2025*
  Title: *Quasi-Normal Modes of Ellis–Bronnikov Wormholes* **On-site**
  Advisor: Dr. Suman Ghosh
  ◦ Investigated quasi-normal modes of Ellis–Bronnikov wormholes under perturbations.
  ◦ Implemented and compared WKB and Prony methods for QNM computation.

Workshops and Summer Courses

KSO Summer School, Indian Institute of Astrophysics [🌐] ***10 days:*** *June 2025*
Kodaikanal, India **On-site**

- Participated in hands-on sessions on astronomy, including data cleaning, and basic analysis techniques.
- Attended lectures on different branches of astronomy including general relativity, planetary science, solar studies, stellar evolution, and active galactic nuclei .

Programming & Computational Skills

- **Programming Languages & Tools:** Python, R (statistics), MATLAB, C, C++, LaTeX, Linux Shell Scripting
- **Scientific Computing (Python):** NumPy, SciPy, Pandas, AstroPy, Astroquery, Numerical Techniques
- **Data Visualization:** Matplotlib
- **Astronomical Data Analysis:** SAO DS9, Aladin Sky Atlas, TOPCAT, IRAF (photometry), DAOPHOT (PSF fitting), CCDLAB (FITS reduction), GLUE
- **Catalogs & Survey Data:** Gaia DR3, Astrosat/UVIT
- **Modeling & Simulations:** Stellar evolution (PARSEC)

Honors and Awards

- **Branch Topper Award, Birla Institute of Technology, Mesra, India**
  Coordinator: Dr. Rajeev Kumar
  ◦ Received a cash prize of INR 15,000 for securing the highest academic rank in the Physics branch during the 2nd–4th semesters, recognizing consistent academic excellence and top departmental performance.

Volunteer Experience

- **Volunteer for Rural Development, Unnat Bharat Abhiyan (UBA)** *August 2024 – Present*
  Coordinator: Dr. Rajeev Kumar
  ◦ Contributing to rural development through Unnat Bharat Abhiyan by engaging in projects aimed at improving infrastructure, sanitation, and education access in villages, while enhancing leadership, problem-solving, and community engagement skills.

Additional Information

Languages:

- Hindi (Native)
- Maithili (Native)
- English (C1)
- German (Beginner)

Interests:

- Wildlife
- Philosophy
- Politics
- Debate